

**SIMATS ENGINEERING
ASSESSMENT TOOLS****COURSE NAME:** Theory of Computation**COURSE CODE:****CSA13**

COURSE OUTCOME COVERED

CO3: Construct regular expressions, and use the pumping lemma and closure properties to analyze regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5**Total Marks:25****Duration : 1 Hour****Experiments**

Code of Conduct:

I Om Sai Reddy T ¹⁹²³⁷²²⁵² (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIMATS ENGINEERING ASSESSMENT TOOLS

- 1 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0A1 \quad A \rightarrow 0A \mid 1A \mid \epsilon$
- 2 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S0 \mid 1S1 \mid 0 \mid 1 \mid \epsilon$
- 3 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S0 \mid A \quad A \rightarrow 1A \mid \epsilon$
- 4 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow 0S1 \mid \epsilon$
- 5 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)
 $S \rightarrow A101A \quad A \rightarrow 0A \mid 1A \mid \epsilon$

Aim:-

To write a C-Program to check whether a given string belongs to the language defined by the CFG:-

$$S \rightarrow 0A1$$

$$A \rightarrow 0A \mid 1A \mid \epsilon$$

This language contains strings that start with 0 & end with 1.

Procedure:-

=* =* =

- ① Read the input string.
- ② Check whether the first character is 0.
- ③ Check whether the last character is 1.
- ④ The characters in between can be either 0 (or) 1.
- ⑤ If all conditions are satisfied, accept the string.
- ⑥ Otherwise, reject the string.

Program:-

=* =* =

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main () {
```

```
    char str [100];
```

```
    int i, valid = 1;
```

```
    printf ("Enter the string: ");
```

```
    scanf ("%s", str);
```

```
    if [str[0] != '0' || str[strlen(str) - 1] != '1']
```

```
        valid = 0;
```

```
for (i=0; i < strlen(s); i++);
```

```
{
    if (isalnum(s[i]) || isalpha(s[i])) {
```

```
        valid = 1;
```

```
    }
    break;
```

```
}
```

```
if (valid)
```

```
    printf("string belongs to the CFG");
```

```
else
```

```
    printf("string does not belong to the CFG");
```

```
return;
```

```
}
```

Result :-

The 'C' program successfully checks whether the given string belongs to the language by $S \rightarrow 0A1$.

②

Aim :-

To write a C program to check whether a given string belongs to the language defined by the CFG:
 $S \rightarrow 000 | 101 | 011 | \dots$

Procedure :-

1. 1. 1.

① Read the input string;

② Compare the first character with the last character.

③ Move towards the centre of the string.

(2)

- If every pair of character is equal, it is a palindrome.
- ⑤ Accept the string if it is a palindrome.
 - ⑥ Otherwise, reject it.

Program:-

=* =* =

```
#include <stdio.h>
#include <string.h>

int main () {
    char str[100];
    int i, j, valid = 1;
    printf ("Enter the string : ");
    scanf ("%s", str);
    i = 0;
    j = strlen(str) - 1;
    while (i < j)
    {
        if (str[i] != str[j])
        {
            valid = 0;
            break;
        }
        i++;
        j--;
    }
    printf ("String belongs to the CFG");
}
```

Result:-

=* =* =

The C- program Successfully executed.

3) Aim:-

To write a C program to check whether a given string belongs to the language defined by the CFG.

$S \rightarrow OSO \mid A$

$A \rightarrow IAI \mid E$

Language: $O^n I^m O^n$.

Procedure:-

=* =* =

- ① Read the input string.
- ② Count the OS at the beginning.
- ③ Count the OS at the end.
- ④ Check whether both counts are equal.
- ⑤ Otherwise, reject it.

Program:-

=* =* =

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main ( )
```

```
char str[100];
```

```
int i, n, left = 0, right = 0, valid = 1;
```

```
printf ("Enter the string: ");
```

```
scanf ("%s", str);
```

```
n = strlen (str);
```

```
while (i >= 0; str[left] == 'O')
```

```
left++;
```

```
i = n - 1;
```

```
{
```

```
if (left != right).
```

2

valid = 0;

for (i = left; i < n - right; i++) {

if (str[i] != '1')

valid = 0;

break;

}

}

Print ("String does not belong to the CFG");

return 0;

}

Result:-
==*==

Thus, C program executed successfully.

4

Aim:-

==*==

To write a C program to check whether a given string belongs to the language defined by the CFG:

$S \rightarrow 0S1 \mid \epsilon$.

Procedure:-

==*==

- 1 Read the input string.
- 2 Count the number of 0's.
- 3 Count the number of 1's.
- 4 Ensure that all 0's occur before all 1's.
- 5 Check whether the number of 0's equal the no. of 1's.
- 6 Otherwise, reject it.

Program:-

= * = * =

```
#include <stdio.h>
#include <string.h>
int main () {
    char str [100];
    int i, zeroes = 0, ones = 0;
    change = 0, valid = 1;
    printf ("Enter the string:");
    scanf ("%s", str);
    for (i = 0; i < strlen(str); i++) {
        if (str[i] == '0' && change == 0)
            zeroes++;
        else if (str[i] == '1') {
            change = 1;
            ones++;
        }
    }
    printf ("string does belongs to the CFG");
    return 0;
}
```

Result:-

Thus, the C program executed successfully,

⑤

Aim:-

= * =

To write a C-program to check whether a given string belongs to the language defined by the CFG:

$S \rightarrow A | 0 | A$

$A \rightarrow 0A | 1A | \epsilon$

procedure :-

* = * = * =

- ① Read the input String.
- ② Search for the substring 101.
- ③ If 101 is found anywhere in the String, accept it.
- ④ Otherwise, reject the string.

Program:-

```
#include <stdio.h>
#include <string.h>
int main () {
    char str [100];
    int i, found = 0;
    printf ("Enter the string:");
    scanf ("%s", str);
    for (i = 0; i < strlen (str) - 2; i++) {
        if (str[i] == '1' &&
            str[i+1] == '0' &&
            str[i+2] == '1')
        {
            found = 1;
            break;
        }
    }
    if (found)
        printf ("string belongs to the CFG");
    else
```

```
printf ("string does not belong to the CFG");
```

```
return 0;
```

```
}
```

Result :-

Thus, the C program Executed Successfully.

SIMATS ENGINEERING ASSESSMENT TOOLS

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary (25 Marks)	Level 3 – Proficient (20-24 Marks)	Level 2 – Developing (10–15 Marks)	Level 1 – Beginning (5 -10 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

91/100

